

SSD_STR Instructions for Use

1 Overview

The full name of STR is Suspend to RAM, which means that the device enters a low-power state and only keeps RAM self-refresh. All device and system states are kept in RAM and all peripherals are suspended.

2. Hardware related

2.1. Matters guaranteed under STR

- a. DDR Power/CPU Power/3.3V Power normal power supply
- b. The power-consuming part of the peripheral equipment is disconnected from the power supply
- c. Pay attention to the default pull-up and pull-down state of the chip GPIO
- d. When SSD20X enters STR Mode, it needs to configure external interrupt wake-up (key/touch, etc.)
- e. When the STR is in standby, the GPIO will keep the setting state applied before the standby. After waking up (reset) the GPIO will return to the default, until the code is run to reset the state.

2.2. STR Mode public power consumption data

	STR Mode				
		电压V	电流mA	功耗mW	Total (mW)
IO	VCC	3.3	4.27	14.091	14.091
DDR	AVDDIO_DRAM	1.8	3.32	5.976	6.678
	AVDDIO_DATA	1.8	0.39	0.702	
CPU	DVDD_DDR	0.9	0.07	0.063	6.363
	VDD	0.9	7	6.3	
	Total				27.132

3. Software related

3.1. APP enables STR

APP hangs:

Execute to `system("echo mem > /sys/power/state");` suspend APP.

Before executing this command, you need to exit all MI modules, and pull down the gpio associated with the module, such as de-initializing disp, Panel, uninstalling wifi ko, turning off the screen backlight, etc.

MI模块有新增API，在模块退出时需要释放模块资源时call对应的DeInitDev接口。如退出DISP模块时，呼叫 MI_DISP_DeInitDev()。

```
MI_S32 SSTAR_DestroyVdec2DispPipe(MI_S32 s32VdecChn, MI_S32
s32DivpChn)
{
    SSTAR_ModuleUnBind(E_MI_MODULE_ID_VDEC, 0, s32VdecChn, 0,
                      E_MI_MODULE_ID_DISP, 0, 0, 0); //DIVP->DISP
    SSTAR_DestroyVdecChannel(s32VdecChn);
    MI_DISP_DisableInputPort(0, 0);

    MI_VDEC_DeInitDev ();
    MI_DISP_DeInitDev ();
}
```

```
    return MI_SUCCESS ;  
}
```

3.2. APP recovery

Wake up through an external interrupt, after waking up, the APP needs to turn on the screen backlight again, load wifi, and re-initialize the Panel and disp.

External interrupt settings:

There is a GPIO in the public board `infinity2m.dtsi` to wake up, and the GPIO can be set to other GPIO specified by the user, or set to the interrupt pin corresponding to the touch screen.

```
370  
371     wakeup:wakeup{  
372         compatible = "sstar,wakeup-gpio";  
373         wakeup_source=<PAD_GPIO0>;  
374     };  
375
```

APP demo is available for reference github.com/aaron201912/UuidSSDPlayer
[UuidSSDPlayer/stdc++/zk_full_str](#) .